

Cooking techniques Classification via Embedding-Based Semantic Clustering of Culinary Actions

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Abstract

Although large-scale recipe repositories have become increasingly available, the procedural semantics embedded within cooking instructions remain poorly structured and difficult to model computationally. Existing food informatics research has largely prioritised ingredients and nutritional analysis, while comparatively limited attention has been given to the semantic organisation of culinary actions. This study addresses this gap by investigating whether modern text-embedding models can recover a meaningful taxonomy of food preparation protocols through unsupervised learning. Culinary action terms were extracted from recipe process annotations and manually organised into fine-grained semantic categories using multiple lexical references. To reduce ambiguity, each protocol was supplemented with context-enriched definitional descriptions before embedding. Multiple representation approaches, including sentence embedding models, domain-adapted transformers, and a distributional baseline, were evaluated using hierarchical agglomerative clustering with cosine distance and average linkage. Clustering quality was assessed using chance-corrected agreement and internal cohesion metrics. Results demonstrate that context-enriched semantic representations substantially improve protocol clustering. A sentence-embedding model achieved the strongest alignment with the curated taxonomy, obtaining an Adjusted Rand Index of approximately 0.53, while several domain-specific baselines exhibited weaker semantic agreement. These findings highlight the importance of contextual semantics in computational modelling of culinary procedures.

Keywords:

1. Introduction

Cooking instructions encode operational knowledge about transforming ingredients through sequences of actions, times, and transformations. Within computational gastronomy, converting these instructions into structured representations supports large-scale analysis of culinary practice and enables model-driven applications that reason over procedures rather than only ingredients. The availability of structured recipe repositories further strengthens this direction by exposing consistent process annotations across cuisines and recipes.

Despite the growth of recipe resources such as RecipeDB, many computational pipelines emphasise ingredient extraction, nutrition linking, or recipe-level similarity, while the semantic organisation of the action vocabulary itself remains comparatively underspecified. In particular,

protocol terms are often polysemous (e.g., “press,” “fold,” “set”) and can denote different operational intents depending on culinary context, which complicates direct modelling. This motivates methodologies that (i) construct a fine-grained, human-interpretable taxonomy of culinary actions and (ii) test whether representation learning can recover that taxonomy via unsupervised structure discovery.

Transformer-based language models, and especially sentence-embedding variants, provide representations optimised for semantic similarity and thus offer a natural basis for clustering action descriptions. However, strong sentence representations are not guaranteed when the input is a single token or short phrase, and evaluation requires a reliable reference partitioning to separate the semantic signal from coincidental geometric structure. Prior work on contextualised language representations and sentence embeddings offers mechanisms for leveraging context and pooling strategies for fixed-length vectors.

The present study makes three contributions. First, a refined semantic taxonomy of culinary protocols is constructed via a multi-source manual labelling procedure. Second, a context-enriched representation strategy is used to embed each protocol through definitional text rather than isolated tokens. Third, multiple embedding models are compared under a common hierarchical clustering and evaluation protocol, with performance quantified using chance-corrected agreement and internal cohesion metrics. The remainder of the paper first details dataset acquisition and ground-truth construction, then describes embedding and clustering methods, and finally reports comparative results and associated limitations.

2. Related Work

Prior work in computational gastronomy and recipe informatics has explored the extraction, representation, and structuring of culinary processes, primarily through verb/action mining, domain-specific embeddings, and clustering techniques applied to ingredients, full recipes, or instructions [1, 2]. However, to the best of our knowledge, no prior study has constructed a fine-grained, human-interpretable taxonomy of individual food preparation protocols (“culinary actions”) and systematically evaluated whether modern text-embedding models can recover this taxonomy through purely unsupervised clustering on context-enriched definitions. The present work addresses this gap while building on established methodological foundations.

Early efforts focused on scalable extraction and analysis of cooking processes from unstructured recipe text. **Chang et al. [20]** introduced RecipeScape, a computational pipeline that mines diverse processes by extracting sequences of cooking verbs/actions from large-scale recipe corpora, computes procedural similarity between recipes, and supports interactive visualisation and comparison of preparation steps. This verb-centric approach demonstrated the value of treating culinary actions as fundamental units for large-scale analysis but relied primarily on rule-based or lexical extraction rather than semantic embeddings or unsupervised clustering.

Subsequent studies advanced semantic representations using domain-adapted embeddings. **Pellegrini et al. [7]** developed Food2Vec (Word2Vec-based) and FoodBERT (transformer-based) models trained on recipe instructions to generate ingredient embeddings, enabling context-free substitution tasks and relation extraction. Similarly, **Morales-Garzón et al. [21]** proposed a domain-specific word-embedding model trained on both ingredient lists and cooking instructions to support unsupervised recipe adaptation (e.g., nutritional or dietary modifications), highlighting the importance of contextual instructions for capturing culinary semantics. **Sözüer et al. [24]** further employed ingredient embeddings combined with hierarchical agglomerative clustering to derive substitutability clusters and generate novel recipe variants.

More recent work has incorporated clustering directly on recipe representations. **Venkataramanan et al. [22]** introduced Ki-Cook, a knowledge-infused multimodal network that clusters recipe embeddings (derived from titles, ingredients, and procedural attributes) using semantic similarity scores, demonstrating improved organisation of cooking representations for retrieval and recommendation. **Ventirozos et al. [23]** applied interactive clustering to recipe instructions for IoT-driven kitchen automation, with manual annotation of sentence-level clusters to detect device-related events.

Domain-specific transformer variants (e.g., **CookBERT [16]**, **BioBERT [6]**, **FoodBERT [7]**) have also been fine-tuned for food-related tasks, often using mean-pooling of token embeddings and cosine-based similarity. These models typically target ingredient-level or full-recipe semantics rather than the isolated protocol vocabulary. Hierarchical agglomerative clustering with average linkage and cosine distance—employed in the present study—has been used in related food-embedding pipelines, often evaluated using metrics like the Adjusted Rand Index [5] or Silhouettes [15], but without the manual fine-grained taxonomy or context-enriched definitional input that mitigates polysemy in action terms.

Collectively, these methodologies establish the feasibility of embedding-based semantic structuring in the culinary domain. The current study extends them by (i) manually constructing a refined protocol taxonomy from **RecipeDB [1]** processes, (ii) enriching single-token protocols with multi-source definitional context from dictionaries [9, 10, 11], and (iii) rigorously comparing general-purpose (**SBERT**) [4] versus domain-adapted embeddings under a unified hierarchical clustering framework. This enables the first quantitative assessment of whether unsupervised representation learning can recover human-interpretable culinary-action categories.

3. Materials and Methods

3.1 Dataset Description

The dataset used in this study was obtained from the Complex Systems Lab at IIIT-Delhi, specifically from the RecipeDB repository available at Foodoscope

(<https://www.foodoscope.com/>). RecipeDB is a large-scale structured dataset containing detailed information about food recipes, including ingredients, nutritional values, and preparation steps. The RecipeDB dataset consists of **118,083 recipes with 24 attributes**, covering recipe metadata, nutritional information, and preparation details. The dataset includes structured fields such as region, preparation time, ingredients, and dietary classifications, enabling diverse analytical perspectives.

The dataset consists of multiple attributes, including *Recipe_id*, *Calories*, *cook_time*, *prep_time*, *servings*, *Recipe_title*, *total_time*, *url*, *Region*, *Sub_region*, *Continent*, *Source*, *img_url*, nutritional components such as *Carbohydrate (g)*, *Energy (kcal)*, *Protein (g)*, *Total lipid (fat) (g)*, and categorical fields like *Utensils* and *Processes*, along with dietary tags such as *vegan*, *pescetarian*, *ovo_vegetarian*, *lacto_vegetarian*, and *ovo_lacto_vegetarian*.

For this study, the ‘processes’ column was utilised as the primary source of data. This column contains textual representations of cooking actions or steps involved in food preparation. From this field, individual cooking actions (verbs) were extracted and treated as fundamental units of analysis, referred to as *food protocols*.

These extracted protocols represent common culinary operations such as cutting, mixing, heating, and preparation actions. A total of 270 unique protocol terms were identified and used for subsequent embedding generation, clustering, and evaluation tasks.

3.2 Protocol Labelling Procedure

Given the set of extracted home-cooked food protocols, the task was to classify each protocol into semantically meaningful subcategories. The entire dataset was labelled manually using a structured and consistent procedure.

To ensure accurate interpretation, the meaning of each protocol word was first analysed using external lexical resources. For each protocol, multiple definitions (typically up to three) were collected from reliable sources such as the *Oxford English Dictionary*, *Cambridge Dictionary*, *Merriam-Webster*, and domain-specific resources like *Food.com*. This multi-source approach helped capture variations in meaning, especially for words with multiple contextual interpretations.

Protocols with clear and consistent meanings across sources were directly assigned to appropriate categories. For ambiguous words with multiple possible interpretations, the most relevant or dominant definition in the context of food preparation was selected. This ensured that labelling remained consistent with culinary usage rather than general language ambiguity.

The labelling process was guided by a semantic understanding of actions performed during cooking, focusing on how each protocol contributes to food preparation.

3.3 Ground Truth Construction

Following the labelling procedure, protocols were organised into a refined set of semantic categories to enable structured analysis and evaluation. Instead of relying on overly broad groupings, the labelling was refined to capture finer distinctions between different types of cooking actions.

The protocols were grouped into semantically coherent subcategories based on their functional role in food preparation. For instance, transformation actions included operations such as *flatten*, *crush*, and *melt*, which alter the physical state or structure of ingredients. Cutting and separation actions covered verbs such as *chop*, *slice*, *dice*, and *tear*, representing different forms of division. Mixing and combination actions included protocols like *mix*, *stir*, *blend*, and *whisk*, which involve combining ingredients.

Thermal processing actions were categorised separately, including operations such as *boil*, *fry*, *bake*, and *steam*, which involve heat-based transformations. Cleaning and preparation actions included *wash*, *rinse*, and *scrub*, reflecting the preprocessing steps applied to ingredients.

In addition to these primary categories, more specialised subcategories were introduced to resolve ambiguity in previously generic labels. For example, rotational or directional actions such as *flip*, *twist*, and *turn* were grouped into a distinct category, while resting or waiting actions such as *rest*, *set*, and *wait* were separated based on their temporal role in preparation. Similarly, opening and separation actions like *peel*, *unwrap*, and *strip* were distinguished from general transformation actions. Other categories included measurement-related actions (*measure*, *scoop*), handling or contact-based actions (*press*, *rub*), control and checking actions (*check*, *test*), and wrapping or covering actions (*wrap*, *cover*).

This refinement addressed the ambiguity present in earlier coarse labels by grouping protocols based on their operational meaning in cooking rather than general linguistic similarity. As a result, semantically similar actions were more consistently categorised, improving the interpretability of clusters and the reliability of evaluation.

The resulting categorisation provides a more expressive and semantically grounded representation of food preparation actions, forming the basis for subsequent clustering and evaluation.

Table 1: Top 10 semantic protocol categories in the manually constructed ground truth

Category	Example Protocols	Number of Protocols
Action	add, place, pour, remove, transfer	38
Add Water	soak, moisten, splash, immerse	11

Cleaning	wash, rinse, scrub, sterilise	9
Cutting	chop, slice, dice, mince, split	22
Flavoring	marinate, season, smoke, dress	13
Light Cooking	simmer, warm, preheat, braise	8
Mixing	mix, stir, combine, blend, fold	13
More Cooking	fry, bake, roast, sauté, broil	12
Reduce Water	evaporate, reduce, drain, blot	12
Shape Change	flatten, knead, melt, roll, soften	34

4. Methodology

This study presents an embedding-based clustering framework for the classification of food preparation protocols. The overall methodology consists of three key stages: (i) transforming protocol words into continuous vector representations using multiple embedding models, (ii) clustering these representations to group semantically similar actions, and (iii) evaluating the clustering performance against a manually constructed ground truth using the Adjusted Rand Index (ARI).

Given a set of protocol terms $\{w_1, w_2, \dots, w_n\}$, each word is first mapped to a high-dimensional embedding vector using a predefined embedding function. These vector representations capture semantic similarities between different cooking actions. The resulting feature space is then used to perform unsupervised clustering, where similar protocol actions are grouped together based on their embedding proximity.

To assess the quality of clustering, the predicted cluster assignments are compared with the refined manual labels. This comparison is performed using ARI, which provides a robust measure of similarity between two partitions while accounting for chance agreement. The methodology is applied consistently across multiple embedding models to analyse their effectiveness in capturing semantic structure within food preparation protocols.

4.1 Embedding Models

To represent food preparation protocols in a continuous vector space, multiple embedding models were employed. These models convert each protocol word into a dense numerical vector that captures its semantic meaning. Formally, each protocol word w_i is mapped to a vector representation $x_i \in R^d$ using an embedding function f :

$$x_i = f(w_i), x_i \in R^d$$

Where d denotes the dimensionality of the embedding space.

In this study, a diverse set of embedding models was used, including both general-purpose and domain-specific representations. These include Sentence-BERT (SBERT), which is designed for capturing semantic similarity at the sentence level, and domain-adapted transformer models such as CookBERT, FoodBERT, BioBERT, and PubMedBERT. Additionally, Doc2Vec was used as a baseline model for comparison.

The use of multiple embedding models allows for a comparative analysis of how different representations capture the semantic relationships between food preparation actions. While general-purpose models aim to capture broad linguistic meaning, domain-specific models are expected to better encode context related to cooking and food preparation. All embeddings were generated independently for each protocol word and subsequently used as input for clustering. Since transformer models produce embeddings for each token, a fixed-length representation for each protocol was obtained using **mean pooling** over the token embeddings. In some models (e.g., BioBERT, PubMedBERT), this pooling was applied explicitly, while in Sentence-BERT-based models, it is handled internally. This aggregation step ensures that each protocol is represented as a single dense vector suitable for downstream clustering.

4.2 Feature Extraction

Rather than representing culinary protocols as isolated lexical units, each protocol was encoded using a context-enriched *merged_text* representation generated during the annotation pipeline. For every protocol term, multiple complementary definitions—typically up to three—were curated from both general lexical dictionaries and domain-specific culinary resources. The collected definitions were subsequently consolidated into a unified textual representation, forming the *merged_text* input used for semantic embedding.

Such contextual aggregation captures multiple semantic dimensions associated with a protocol, thereby reducing lexical ambiguity and preserving nuanced procedural meanings across diverse culinary contexts. Consequently, the embedding models learn semantically richer and functionally grounded representations, enabling protocols to be organised according to their operational role in food preparation rather than superficial lexical similarity alone.

4.3 Model Methodology

The food protocol classification framework was implemented using a multi-model embedding and clustering pipeline designed to compare semantic representations across general-purpose, biomedical, and food-domain language models. Each protocol term was first represented using

the **merged_text** field, which combined the original cooking protocol word with its manually collected dictionary definitions and contextual descriptions. This preprocessing step ensured that each protocol preserved both lexical meaning and culinary usage before embedding generation. The experimental framework was implemented using a combination of established scientific computing and deep learning libraries. Numerical computation and vector operations were performed using *NumPy*, while *Pandas* was employed for structured data handling and preprocessing. Model inference and tensor-based computations were conducted using *PyTorch*. Pretrained transformer architectures were accessed through the *Hugging Face Transformers* ecosystem, whereas sentence-level embedding generation was facilitated using the *SentenceTransformers* library. The distributional baseline model was implemented using *Gensim*. Clustering, similarity computation, and evaluation metrics were computed using *scikit-learn* and *SciPy*. Additionally, *UMAP-learn* was used for nonlinear dimensionality reduction and low-dimensional visualisation of semantic embedding spaces, while *Matplotlib* and *Seaborn* were employed for graphical analysis and result visualisation.

Table 2: Language model architectures and embedding generation configurations used for protocol representation and comparison.

Model	Details
SBERT	Pretrained checkpoint: all-mpnet-base-v2 from SentenceTransformers. Generated 768-dimensional normalised sentence embeddings using batch size 32 . Cosine similarity was computed directly on normalised vectors.
BioBERT	Primary checkpoint: pritamdeka/biobert-base-cased-sentence-transformer . Fallback checkpoint: dmis-lab/biobert-base-cased-v1.1 . Mean pooling was applied over the final hidden layer using attention masks. Produced 768-dimensional embeddings with batch size 32 (or 16 in fallback mode).
PubMedBERT	Checkpoint: microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract . Tokenisation used a maximum sequence length of 512 . Mean pooling was applied over token embeddings. Produced 768-dimensional embeddings using batch size 16 .
RecipeBERT	Checkpoint: alexandseo/RecipeBERT . Domain-specific transformer trained on recipe text. Mean pooling over the final hidden states was used to generate 768-dimensional embeddings with a batch size of 16 .
FoodBERT	Implemented using alexandseo/RecipeBERT as a food-domain semantic encoder. Mean-pooled token representations were used. Generated 768-dimensional embeddings with batch size 16 .

CookBERT	Cooking-domain baseline initialised from bert-base-uncased . Mean pooling over contextual token embeddings. Produced 768-dimensional embeddings with batch size 16 .
Doc2Vec	Trained using Gensim Doc2Vec . Parameters: vector size = 300 , window = 8 , minimum word count = 2 , epochs = 40 , workers = 4 , random seed = 42 . Final protocol representations were obtained from document vectors.

Transformer-based models produced contextual token embeddings from the final hidden layer. These token vectors were converted into sentence-level protocol embeddings using mean pooling over valid tokens:

$$E = \frac{\sum_{i=1}^n h_i \cdot m_i}{\sum_{i=1}^n m_i}$$

Where h_i denotes the hidden representation of token i , and m_i is the corresponding attention mask.

4.4 Clustering Approach

To group semantically similar food preparation protocols, a clustering framework was applied over the extracted embedding representations. This process was performed independently for multiple embedding models, including Sentence-BERT (SBERT), BioBERT, PubMedBERT, FoodBERT, CookBERT, and Doc2Vec, to enable comparative analysis of their effectiveness. The number of clusters was fixed at **30**, since the manually curated ground-truth protocol taxonomy contained approximately 30 semantic protocol categories. Aligning the clustering output with this reference enabled direct comparison between unsupervised clusters and expert-defined labels.

Given the embedding matrix $X = \{x_1, x_2, \dots, x_n\}$, where each $x_i \in R^d$, Pairwise semantic similarity between protocol representations was computed using **cosine similarity**:

$$sim(x_i, x_j) = \frac{(x_i * x_j)}{(\|x_i\| \|x_j\|)}$$

This similarity measure captures the angular closeness between vectors in high-dimensional space and is well-suited for semantic representations generated by embedding models.

To convert similarity into a distance metric suitable for clustering, the following transformation was applied:

$$d(i, j) = 1 - sim(x_i, x_j)$$

Clustering was performed using **Hierarchical Agglomerative Clustering (HAC)**, a bottom-up approach that starts with each protocol as an individual cluster and iteratively merges the closest pairs of clusters based on a defined distance criterion.

The clustering process used **average linkage**, where the distance between two clusters A and B is defined as the average of all pairwise distances between elements of the clusters:

$$D(A, B) = \frac{1}{|A| \cdot |B|} \sum_{x_i \in A} \sum_{x_j \in B} d(i, j)$$

Average-linkage hierarchical agglomerative clustering was employed to construct semantic groupings of culinary protocols. Under this formulation, the inter-cluster distance is computed as the mean cosine distance across all pairwise embedding combinations between two clusters. The adoption of average linkage provides a balanced estimation of cluster similarity while reducing sensitivity to extreme observations relative to single-linkage and complete-linkage strategies. Such characteristics are particularly advantageous in dense contextual embedding spaces, where average linkage yields comparatively more stable and semantically coherent cluster structures.

4.5 Evaluation Metrics

The quality of clustering was evaluated by comparing the predicted cluster assignments with the manually constructed ground-truth semantic labels using three complementary external evaluation metrics: **Adjusted Rand Index (ARI)**, **Normalised Mutual Information (NMI)**, and **V-measure**. These metrics quantify how closely the automatically generated clusters match the manually annotated protocol categories.

4.5.1 Adjusted Rand Index (ARI)

The **Adjusted Rand Index (ARI)** measures the similarity between two clustering while correcting for agreements that may occur by chance. It is defined as:

$$ARI = \frac{\sum_{ij} \frac{n_{ij}}{2} - [\sum_i \frac{a_i}{2} \sum_j \frac{b_j}{2}] / \frac{n}{2}}{\frac{1}{2} [\sum_i \frac{a_i}{2} + \sum_j \frac{b_j}{2}] - [\sum_i \frac{a_i}{2} + \sum_j \frac{b_j}{2}] / \frac{n}{2}}$$

Where:

n_{ij} : overlap between clusters

a_i : size of cluster in U

b_j : size in V

n: total samples

Among all models, **SBERT achieved the highest ARI score of 0.5360**, indicating the strongest alignment between predicted clusters and the manually defined semantic labels.

4.5.2 Normalised Mutual Information (NMI)

The **Normalised Mutual Information (NMI)** measures the amount of shared information between the predicted and manual cluster assignments. It is defined as:

$$NMI(U, V) = \frac{I(U; V)}{\sqrt{H(U) * H(V)}}$$

where:

$I(U; V)$ is the mutual information between clustering UUU and VVV

$H(U)$ and $H(V)$ are the entropies of the two clustering

The **SBERT model achieved the highest NMI of 0.7144, indicating it preserved the most semantic information relative to the manual labels.**

4.5.3 V-measure

The V-measure evaluates clustering quality as the harmonic mean of homogeneity and completeness, where:

Homogeneity: each predicted cluster contains protocols from a single manual category

Completeness: all protocols of one manual category appear in the same predicted cluster

$$\text{It is computed as: } V = 2 * \frac{\text{Homogeneity} * \text{Completeness}}{\text{Homogeneity} + \text{Completeness}}$$

The highest V-measure was again obtained by SBERT (0.7140), confirming that its clusters were both internally consistent and semantically complete.

4.5.4 Computation Procedure

The evaluation pipeline was designed to systematically compare model-generated protocol clusters against the manually curated semantic taxonomy. The overall computation procedure consisted of four major stages: ground-truth preparation, predicted cluster extraction, label alignment, and quantitative evaluation.

Step 1: Ground-Truth Label Construction

The manually annotated semantic reference dataset was loaded from *protocol_manual_classification_updated.csv*. To ensure lexical consistency during matching, all protocol terms were standardised through lowercase conversion and whitespace normalisation. Semantic categories provided in the *REMARK* field were subsequently transformed into categorical numerical identifiers using label encoding. These encoded identifiers served as the ground-truth cluster labels for downstream evaluation.

Step 2: Predicted Cluster Acquisition

For each embedding model, clustering outputs were stored as independent cluster files (*cluster_1.csv*, *cluster_2.csv*, *cluster_3.csv*, etc.), where each file contained the protocol terms assigned to a specific cluster. The cluster index extracted from the corresponding filename was treated as the predicted cluster label generated by the unsupervised clustering framework.

Step 3: Cluster–Label Alignment

Protocol terms obtained from the predicted clusters were matched against the manually annotated reference dataset. Only protocol terms common to both datasets were retained to ensure a valid comparative evaluation. The resulting aligned representation consisted of:

manual_cluster: manually assigned semantic category labels

predicted_cluster: model-generated cluster assignments

Step 4: Quantitative Evaluation

Clustering quality was evaluated using both external agreement metrics and internal cluster validation measures:

$$ARI = AdjustedRandIndex(manual_cluster, predicted_cluster)$$

$$NMI = NormalizedMutualInformation(manual_cluster, predicted_cluster)$$

$$V = V - Measure(manual_cluster, predicted_cluster)$$

$$S = SilhouetteScore(embeddings, predicted_cluster)$$

The Adjusted Rand Index (ARI) measures clustering agreement corrected for random assignment, where 1.0 indicates perfect correspondence, 0.0 reflects random clustering behaviour, and negative values denote agreement below random expectation. Normalised Mutual Information (NMI) and V-measure quantify the mutual dependence and homogeneity–completeness balance between predicted and reference labels, with values approaching 1.0 indicating strong semantic agreement. Internal cluster cohesion was assessed using the Silhouette Score, where values near +1 represent well-separated clusters, values around 0 indicate overlapping cluster boundaries, and negative values suggest incorrect cluster allocation.

5. Embeddings visualization methodology

To examine the geometric structure of protocol representations generated by each embedding model, the high-dimensional embedding vectors were projected into lower-dimensional spaces using three complementary visualisation methods: **Principal Component Analysis (PCA)**, **Uniform Manifold Approximation and Projection (UMAP)**, and **t-distributed Stochastic Neighbour Embedding (t-SNE)**. These techniques were applied independently to the

embeddings produced by all evaluated models, including **SBERT, BioBERT, PubMedBERT, RecipeBERT, FoodBERT, CookBERT, and Doc2Vec**.

Before visualisation, each embedding matrix $X \in R^{n \times d}$ was standardised using z-score normalisation:

$$x_{scaled} = \frac{X - \mu}{\sigma}$$

where:

n is the number of protocol terms,
d is the embedding dimension,
 μ is the mean of each feature,
 σ is the standard deviation of each feature.

Standardisation ensured that all embedding dimensions contributed equally during dimensionality reduction.

5.1 Principal Component Analysis (PCA)

PCA was first used to evaluate the variance captured by the principal components of each embedding space. The cumulative explained variance was computed as:

$$C_k = \sum_{i=1}^k \lambda_i$$

where:

λ_i denotes the explained variance ratio of the i th principal component,
 C_k is the cumulative variance retained by the first k components.

This analysis helped determine how efficiently each model compressed semantic information. Two-dimensional PCA projections were then generated using:

$$Z_{PCA} = X_{scaled} W \quad \text{where } W \text{ contains the first two principal eigenvectors.}$$

The 2D projections were colored according to the manually assigned semantic labels to assess visual separation among protocol categories.

5.2 UMAP Projection

To preserve local neighbourhood structure within the high-dimensional embedding space, Uniform Manifold Approximation and Projection (UMAP) was employed for nonlinear dimensionality reduction and low-dimensional visualisation. The projection was generated using a two-dimensional latent space ($n_components = 2$), with the number of nearest neighbours set to 15 to balance local and global structural preservation. A minimum distance parameter of 0.1 was used to maintain moderate cluster compactness, while cosine distance was selected as the similarity metric to ensure consistency with the embedding-based semantic similarity

framework. For reproducibility, all UMAP transformations were initialised using a fixed random seed of 42.

UMAP constructs a weighted graph in the original space and optimises a low-dimensional representation that preserves local manifold structure:

$$Z_{UMAP} = f_{UMAP}(X_{scaled})$$

Where f_{UMAP} denotes the nonlinear transformation.

This allowed semantically similar cooking actions to appear closer together while preserving cluster boundaries.

5.3 t-SNE Projection

To further examine local semantic organisation within the embedding space, t-distributed Stochastic Neighbour Embedding (t-SNE) was employed for nonlinear dimensionality reduction and visualisation. The transformation was performed in a two-dimensional latent space ($n_{components} = 2$) using a perplexity value of 30 to balance local neighbourhood sensitivity and global manifold structure. Principal Component Analysis (PCA) was used for initialisation to improve convergence stability, while the learning rate was configured using the adaptive *auto* setting. All experiments were initialised with a fixed random seed of 42 to ensure reproducibility of the projected embedding distributions.

t-SNE models pairwise similarities between samples in both high and low dimensions by minimising the Kullback–Leibler divergence:

$$KL(P||Q) = \sum_i \sum_j P_{ij} \log \frac{P_{ij}}{Q_{ij}}$$

where:

P_{ij} is a similarity in the original space,

Q_{ij} is similarity in the projected space.

This projection emphasised fine-grained local semantic clusters among protocol actions.

6. Results

The correlation heatmap provides important insights into the relationships between recipe preparation attributes and nutritional composition variables within the dataset. A moderate positive correlation is observed between cook time and prep time, indicating that recipes requiring extensive preparation generally also involve longer cooking durations. Nutritional attributes such as Energy (kcal), Protein (g), Total lipid (fat) (g), and Carbohydrate, by difference

(g) exhibit comparatively stronger positive correlations with one another, reflecting the interconnected contribution of macronutrients toward the overall caloric content of recipes. In particular, Energy (kcal) demonstrates a strong positive association with Total lipid (fat) (g), suggesting that fat content significantly influences the caloric density of recipes. Protein content also shows a positive relationship with energy values, indicating that protein-rich recipes tend to contain higher energy levels. In contrast, preparation-related variables display relatively weak correlations with nutritional attributes, implying that recipes with longer preparation or cooking durations are not necessarily more calorie-dense or nutritionally richer. Additionally, the absence of extremely high correlations among most variables suggests low multicollinearity within the dataset, indicating that the features contribute distinct and meaningful information for downstream embedding generation, clustering, and semantic analysis tasks. Overall, the heatmap highlights the diversity and heterogeneity of the recipe dataset, supporting its suitability for exploratory data analysis and machine learning-based clustering approaches.

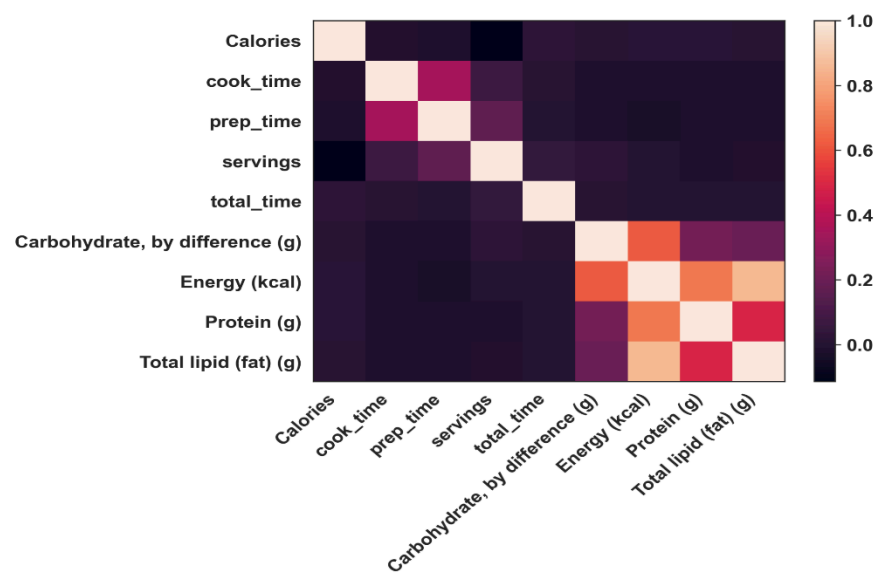


Figure 1: Correlation heatmap showing relationships among recipe preparation attributes and nutritional features in the dataset.

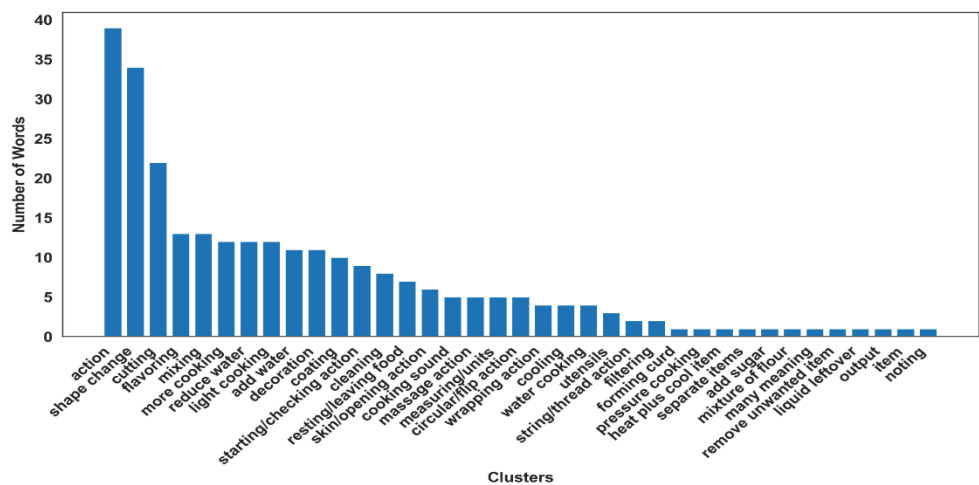


Figure 2: Distribution of manually defined semantic protocol categories used for ground-truth construction.

The cluster distribution plot shows the frequency of protocols across the refined semantic categories defined during ground-truth construction. A clear imbalance is observed, with categories such as *action*, *shape transformation*, and *thermal processing* dominating the dataset, while several other categories contain relatively fewer instances. This skewed distribution reflects the natural prevalence of certain cooking operations in real-world recipes.

From a modelling perspective, such an imbalance can influence clustering behaviour, leading to denser, more coherent clusters for high-frequency categories, whereas sparsely represented categories may exhibit weaker structural cohesion. This observation is important for interpreting clustering performance, as it highlights potential biases in representation and emphasises the need for careful evaluation of minority categories.

Table 3: Quantitative evaluation of semantic clustering performance across embedding models using **Silhouette Score**, **Adjusted Rand Index (ARI)**, **Normalised Mutual Information (NMI)**, and **V-measure**.

Model	Silhouette	ARI	NMI	V-measure
SBERT	0.1655	0.5360	0.7144	0.7140
CookBERT	0.1228	0.2810	0.6220	0.6212
BioBERT	0.0932	0.1678	0.5390	0.5379
RecipeBERT	0.0586	0.1550	0.5491	0.5458
FoodBERT	0.0587	0.1550	0.5491	0.5458
PubMedBERT	0.0212	0.0686	0.4388	0.4242
Doc2Vec	0.6333	-0.0036	0.2286	0.1777

Table 3 presents the comparative performance of all embedding models across four clustering evaluation metrics: Silhouette Score, Adjusted Rand Index (ARI), Normalised Mutual Information (NMI), and V-measure. Among all models, SBERT achieved the best overall performance with an ARI of 0.5360, NMI of 0.7144, and V-measure of 0.7140, indicating the strongest agreement between the generated clusters and the manually defined protocol

categories. Its silhouette score of 0.1655 further suggests that the clusters were reasonably well separated in the embedding space.

The domain-specific CookBERT model showed the second-best performance, with an ARI of 0.2810 and NMI of 0.6220, demonstrating that cooking-oriented semantic information improved clustering quality compared to other specialised language models. BioBERT, RecipeBERT, and FoodBERT produced moderate performance, indicating partial semantic understanding of food preparation actions but weaker alignment with the ground-truth taxonomy.

Although Doc2Vec achieved the highest silhouette score (0.6333), its negative ARI (-0.0036) revealed that the resulting clusters did not correspond to the manually assigned semantic labels. This suggests that while the clusters were internally compact, they lacked meaningful semantic structure. PubMedBERT produced the weakest performance among transformer-based models, likely due to the mismatch between biomedical pretraining and culinary protocol terminology.

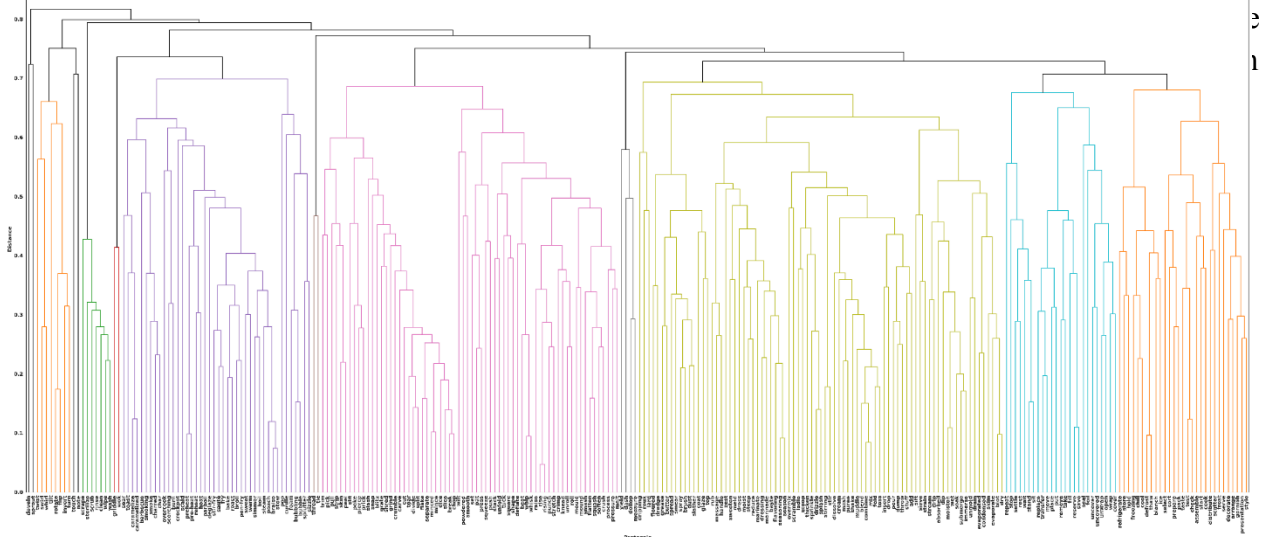


Figure 3: Hierarchical clustering dendrogram of food preparation protocols generated using SBERT embeddings and average-linkage agglomerative clustering.

The dendrogram illustrates the hierarchical organisation of food preparation protocols based on SBERT-derived embeddings. Semantically similar protocols are merged at lower linkage distances, forming tightly connected subclusters, whereas semantically distant actions are merged only at higher levels of the hierarchy. The presence of well-defined branching structures indicates that the embedding space effectively captures meaningful semantic relationships among culinary actions. Furthermore, the hierarchical nature of the clustering enables interpretation at

multiple levels of granularity, revealing both fine-grained distinctions (e.g., variations of cutting actions) and broader groupings (e.g., transformation versus thermal operations). This supports the suitability of hierarchical clustering for modelling structured semantic relationships in food protocols.

The cumulative explained variance plot obtained from Principal Component Analysis (PCA) provides insight into the intrinsic dimensionality of the SBERT embedding space. A substantial proportion of the variance is captured by the first few principal components, after which the curve exhibits diminishing returns, suggesting that semantic information is concentrated in a few dominant representations, which encode the majority of the variance.

Figure 4: Cumulative explained variance obtained from Principal Component Analysis (PCA) of SBERT embeddings.

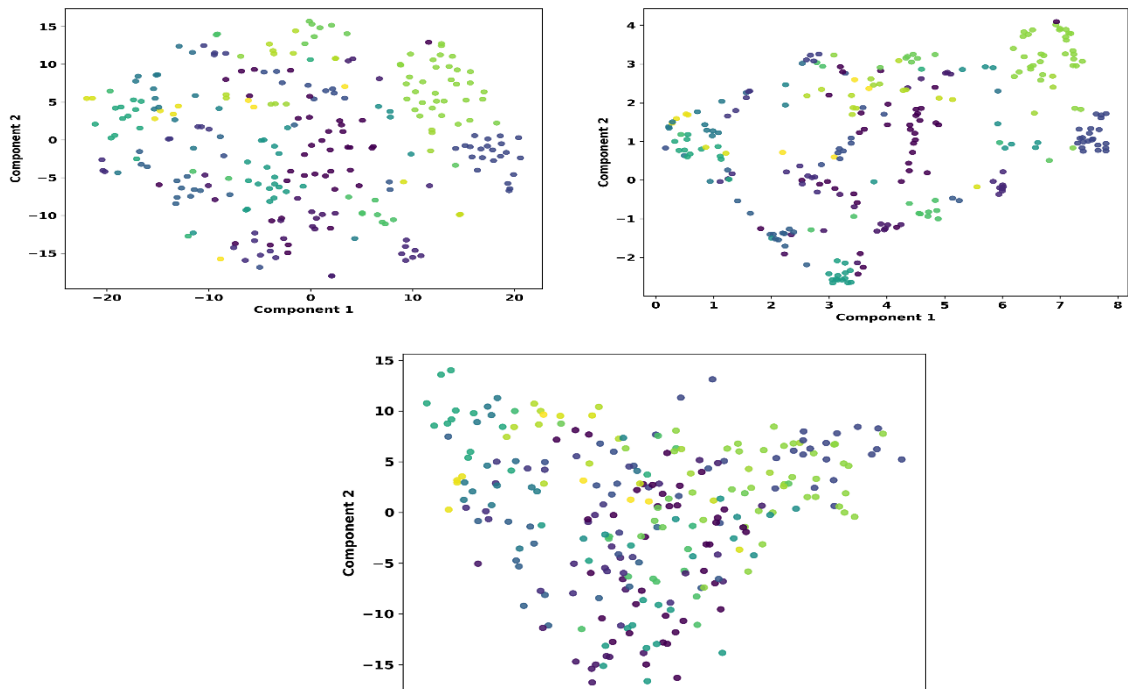


Figure 5: Comparative low-dimensional visualisation of SBERT protocol embeddings using (a) t-SNE, (b) UMAP, and (c) PCA, illustrating the semantic organisation of food preparation protocols across different dimensionality reduction techniques.

The low-dimensional projections presented in Figure 5 provide complementary perspectives on the semantic structure captured by SBERT embeddings for culinary protocols. The t-SNE representation (Figure 5a) reveals the emergence of several localised clusters, demonstrating that semantically related cooking protocols are embedded in proximity. However, partial overlap among clusters is also evident, reflecting the contextual and functional similarities shared across certain culinary actions. Such overlap highlights the inherent semantic continuity and ambiguity present in natural language descriptions of food preparation processes.

In contrast, the UMAP projection (Figure 5b) exhibits comparatively denser and more distinctly separated clusters, indicating superior preservation of both local and global topological relationships within the embedding space. The PCA projection (Figure 5c), while useful for capturing directions of maximum variance through linear transformation, displays a more diffuse distribution with less pronounced cluster boundaries. This observation underscores the limitations of linear dimensionality reduction in modelling complex semantic relationships and further justifies the use of non-linear techniques such as t-SNE and UMAP for exploratory analysis of semantic embedding spaces.

The semantic cluster structure derived from Sentence-BERT (SBERT) embeddings was systematically investigated using multiple dimensionality reduction techniques, including PCA, t-SNE, and UMAP, alongside quantitative evaluation through the Adjusted Rand Index (ARI). Among the evaluated embedding representations, SBERT demonstrated the strongest clustering capability, achieving an ARI score of 0.53, which indicates a substantial correspondence between the learned semantic groupings and the manually curated ground-truth protocol categories. These findings highlight the effectiveness of SBERT embeddings in capturing semantically meaningful relationships within culinary protocol data.

The cumulative explained variance profile obtained through Principal Component Analysis reveals that a considerable proportion of the embedding variance is retained within the first approximately 100 principal components, followed by a gradual saturation trend. This suggests that, despite the inherently high-dimensional nature of SBERT embeddings, a significant amount

of semantic information resides within a comparatively lower-dimensional manifold. Nevertheless, the absence of sharp variance concentration within only a few components indicates that semantic knowledge is broadly distributed across multiple latent dimensions, reflecting the richness and contextual complexity encoded within the embedding space.

7. Conclusion & Discussion

The superior performance of SBERT can be attributed to its training objective, which is specifically designed to capture semantic similarity between textual inputs. Unlike domain-specific models such as BioBERT or PubMedBERT, which are optimised for biomedical contexts, SBERT is better aligned with general language semantics, making it more effective for its representing diverse culinary actions. Additionally, the use of merged textual definitions further enhances its ability to capture contextual meaning.

8. Code and Data Availability: <https://github.com/Harshvishwa123/BTP-8TH-SEM>

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